Rate-Dependent Necking Onset in Cold-Rolled AA6082 Sheet

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Abstract

Uniaxial tensile tests on cold-rolled AA6082 sheet were performed across four strain rates from 10^{-3} to 10^{2} s^{-1} to characterise the onset of diffuse necking. The critical strain at necking onset, identified from digital image correlation strain fields, decreases by approximately 18% between the quasi-static and highest dynamic rate tested. A modified Considère criterion incorporating a rate-sensitivity term is shown to predict the observed onset strains within 6% across the full rate range.

1 Introduction

Diffuse necking marks the practical ductility limit of a sheet metal forming operation, and its rate dependence is of direct relevance to crash and high-speed forming simulations. This short communication reports necking onset strains for AA6082-T6 sheet across four decades of strain rate.

2 Experimental

Dogbone specimens machined from 2 mm cold-rolled AA6082-T6 sheet were tested in uniaxial tension on a servo-hydraulic test rig instrumented with digital image correlation. Necking onset was identified as the strain at which the strain-rate field first develops a localised peak exceeding twice the nominal applied rate.

3 Results

$$\varepsilon_{\mathrm{neck}}(\dot{\varepsilon}) = \varepsilon_0 \left(1 - \beta \log_{10} \frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right) \tag{1}$$

fits the four measured onset strains with $\varepsilon_0 = 0.184$ and $\beta = 0.041$, giving a maximum residual of 6% across the tested range.

4 Conclusion

The modified Considère criterion above captures the observed rate sensitivity of necking onset in cold-rolled AA6082 sheet without requiring a full crystal-plasticity treatment, and is suitable for direct use in sheet-forming process models.